Web application development M1-2025 - Projet (EN)

Introduction

This project aims to confirm everything seen in the web application development course from M1-S2-2025.

Due date is on the **13th of March, 23h59**.

The repository is to return, with the names of the group's members, to **gerald.gallet@ext.junia.com**.
If the repository is private, please invite @GeraldGallet on Github on it.

You will work in **groups of 4 persons**, and the grade will be common.

You are free to split the work however you want : half on front, half on back, or each person has his/her feature, anything that works for you.

You will return a Github repository.

Cleanliness of the commit history will be accounted for in the grade.

You all start from a common codebase - that matches the features we developed during class - and you will add as much feature as you can.

The quality of the produced code is more important than the number of delivered features.

You implement the features in the order you want. Think about this order before starting, in order to be able to deliver incremental changes in a clean way.

The codebase can be found here : https://github.com/GeraldGallet/m1-s1-2025-project

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Technical stack

The technical stack of the project is as follows

- Back-end
  - Typescript

    - NestJS
    - TypeORM
    - SQLite
    - API REST (no GraphQL)
- Front-end
    - Typescript
    - React
    - Vite
    - Ant Design
    - @tanstack/react-router

No deviation from the class's stack will be tolerated.

You will follow the architecture seen in class : layer separation, organization by domain.

The ESLint rules shall be respected. If a file opens in red on VSCode, this will mean less points.

Every `any` (even ignored), means less points.

The following types will be explicited :

- function parameters
- function return
- generic types (ex. `useState<string>()`)

No design was prepared : have fun.

The only rule is to use the requested component when specified : if we ask for a modal, a modal has to be used.

Features

Here is the list of features to implement :

Data model

You will need to add a system to manage clients.

A client has

- a last name
- a first name
- an optional email adress
- an optional picture → you can store the link to a photo, instead of the photo itself

On books and authors, you will add a picture.

You must implement a sales system : we will be able to record that a client X has bought a book Y at a Z date.

Clients list

- A page `/clients` lets me see the list of clients.
  - A list is displayed
    - I see the first and last name of the client
  - From this list, I can delete a client
    - A modal asks me to confirm my choice
  - When I click on a client, I navigate to the details' page of this client
  - Is see, for each client, the number of books he/she bought
- A button lets me open a modal to create a client.

Details of a client

A page `/clients/$clientId` lets me see the details of a client.

- Information of the client are displayed.
- I can update information of the client on the page.
- I see the list of books the client has bought
  - I see the name of the book, the author, and the sales date
  - When I click on a book, I navigate to this book's details page.

Books list

A page `/books` lets me see the list of books.

- A list is displayed.
- I can delete a book from this list : there is a modal asking me to confirm my choice.
- I see the number of clients that bought this book.

Details of a book

A page `/books/$bookId` lets me see the details of a book.

- I see all information of a book.
- I can update the information on the book.
- I see the list of clients that bought this book.
- I have a button that opens a modal used to record a sale
  - This modal lets me chose a client, a date, and save the sale.

Authors list

A page `/authors` lets me see the list of authors.

- I see the list of authors
- I can delete an author from this list : there is a modal asking me to confirm my choice.
- For each author, I see the number of book he/she wrote
- There is a button that opens a modal to create an author.
- When I click on an author, I navigate to the author's details page.

Details of an author

A page `/authors/$authorId` lets me see the details of an author.

- I see all information of the author.
- I can update the information of the author.
- I see the list of books written by the author.
  - When I click on one of the books, I navigate to the book's details page.
- I see the average number of sales of this author's books.

Navigation & Layout

Each domain has a breadcrumb : https://ant.design/components/breadcrumb

When I'm on the books' page, this breadcrumb only displays "Books".
When I am in the details of a book, "Books" → "<name of the book>"
When I click on "Books", I return on the books' page.

This, for all resources.

The menu lets me navigate

- to the home page

- to the list of each resource